# 1

# A HISTORICAL OVERVIEW OF M-LEARNING

## *Toward Learner-Centered Education*

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A consideration of all the various historical and cultural events that have led to mobile learning (m-learning) would trace back through history far beyond the invention of Gutenberg's printing press and the influence of the Industrial Revolution. Although it needs to be acknowledged that these events have enabled the mobile age to reach where it is today, this chapter looks more specifically into recent history, starting when the mobile technological epoch began to take shape. In order to explain the history, *mobile* and *learning* have been separated, before I explicitly detail the interconnections for what has now become this young field of m-learning. The chapter will begin by explicating the philosophical, pedagogical, and conceptual underpinnings regarding learning, particularly toward learner-centered pedagogies. This will be followed by a discussion of the technology, covering the evolution of the hardware/software, its adoption into society, and how these technological advancements have led to today's new affordances for learning.

## DEFINING MOBILE LEARNING

At this time, there is no definitive definition of m-learning. If terms such as *distance education* are any indication, there probably will not be a lasting definition of *m-learning* for a long time to come. In January 2005, Laouris and Eteokleous (2005) reported receiving 1,240 items when searching Google for the terms + "mobile learning" + "definition"; remarkably, when they conducted the same search in June 2005, Google provided 22,700 items. So, it appears that 2005 was the year in which m-learning became a recognized term.

In m-learning's relatively short existence, scholars and practitioners have attempted to define it. An early definition of m-learning was simply the use of a palm as a learning device (Quinn, 2000; Soloway et al., 2001). Since then, deep debates have been ongoing as to which attributes should be included in a definition of m-learning (e.g., Laouris & Eteokleous, 2005; Sharples, Taylor, & Vavoula, 2007; Traxler, 2009), and, from a study of the literature, it appears that pedagogy, technological devices, context, and social

interactions are the four central constructs. For example, O'Malley et al. (2003) defined m-learning as, "Any sort of learning that happens when the learner is not at a fixed, predetermined location, or learning that happens when the learner takes advantage of learning opportunities offered by mobile technologies." In O'Malley et al.'s definition, the initial focus is contextual, although closely followed by pedagogies and technologies.

Traxler's (2005) early definition, "any educational provision where the sole or dominant technologies are handheld or palmtop devices," was a good example of a definition centered on the technology. Many early definitions were criticized for taking a technocentric approach (Traxler, 2010). One issue that is agreed upon by academics and practitioners is that further research is necessary to better understand the field of m-learning (Goh & Kinshuk, 2006), which will undoubtedly lead to many further changes to the definition of m-learning.

Sharples et al. (2007) defined m-learning as, "The process of coming to know through conversations across multiple contexts amongst people and personal interactive technologies" (Sharples et al., 2007, p. 4). Although this definition included the four central constructs of m-learning (namely, pedagogy, technological devices, context, and social interactions), the definition is somewhat confusing and ambiguous. For example, the word *conversations* is used early in the definition, which highlights the importance of this word to the definition, and, yet, the definition of conversation is "Oral exchange of sentiments, observations, opinions or ideas" ("Conversation," *Merriam–Webster Dictionary*, 2011). Does this, then, mean that m-learning is centered round verbal communication? Sharples et al.'s definition was written for an article that highlighted conversational theory, and, although they may have intended for conversation to be interactions in general, a word has been chosen that connotes simply *oral interactions.*

Therefore, for the purpose of the chapter, and this book at large, the author of this chapter and the editors of this book (Crompton, Muilenburg, and Berge) have modified Sharples et al.'s (2007) definition. This new definition includes the four central constructs of m-learning, but the wording has been chosen to reduce ambiguity, and additional punctuation has also been included for clarity. Therefore, Crompton, Muilenburg, and Berge's definition for m-learning is "learning across multiple contexts, through social and content interactions, using personal electronic devices."

To be clear, the word "context" in this definition encompasses m-learning that is formal, self-directed, and spontaneous learning, as well as learning that is context aware and context neutral. In other words, the learning may be directed by others or by oneself, and it can be an unplanned, spontaneous learning experience; learning can happen in an academic setting, or any other non-academic setting; and the physical environment may or may not be involved in the learning experience.

Therefore, m-learning can occur inside or outside the classroom, participating in a formal lesson on a mobile device; it can be self-directed, as a person determines his or her own approach to satisfy a learning goal; or spontaneous learning, as a person can use the device to look up something that has just prompted an interest. The environment may be part of the learning experience (e.g., scanning codes to obtain further information about an exhibit in a museum), or the environment may have a neutral role in the learning experience (e.g., reading articles from the Web while traveling on the bus).

## PEDAGOGICAL SHIFTS IN LEARNING

Throughout history, learning has been of paramount importance in all cultures. In simple terms, learning is essential to personal and professional survival, and a culture's pedagogical choice is often driven by social behavior, expectations, and values. For example, Western pedagogies during the 1930s did not encourage autonomy and self-direction. A student was to learn facts without question. Even into the 1950s, pedagogies typically emulated the *tabula rasa* approach, teaching the students as though they were empty vessels waiting for the teachers to impart knowledge.

Learners today are viewed very differently: students are encouraged to be active in their own learning, to be self-thinking and active consumers of knowledge. Historical components that form a background to this cultural and societal pedagogical shift include: reactions to behaviorism, linguistic pragmatism, minority rights movements, increased internationalism, and wider access to education (Gremmo & Riley, 1995). Shifts in educational philosophy have been led by calls for change toward active learnership. Piaget (1929) pioneered the transition from the *tabula rasa* view of young learners, to instead positing learners with complex cognitive structures, seeking environmental stimulation to promote intellectual development.

Building from Piaget's (1929) position toward cognitive theories of learning, Bruner (1966) added that learners use current and past knowledge during the active learning process. Soon afterward, the learner-centered pedagogical epoch commenced.

### *Discovery Learning in the 1970s*

Learning the heuristics of discovery through active participation was Bruner's (1966) recipe for increasing intellectual potency. He believed students are more likely to remember concepts they deduce on their own. This philosophy led to the discovery-learning movement, with the focus on how students acquire, retain, and recall knowledge, a transition from the behaviorist stimulus–response approach. Unfortunately, technology in schools was generally lagging behind instructional pedagogies; the few schools that had computers in the 1970s utilized behavioristic computer-assisted learning programs (Lee, 2000). The World Wide Web (WWW) would have been a great learning support to discovery learning, although only a small number of people had Internet access until the 1990s.

### *Constructivist Learning in the 1980s*

Constructivism is an epistemic belief about how students learn. Following Piaget's (1929), Bruner's (1966), and Jonassen's (1999) educational philosophies, constructivists proffer that knowledge acquisition develops through interactions with the environment. During the 1980s, the development and distribution of multimedia personal computers offered such an interactive method of learning. "The computer was no longer a conduit for the presentation of information: it was a tool for the active manipulation of that information" (Naismith, Lonsdale, Vavoula, & Sharples, 2004, p. 12).

### *Constructionist Learning in the 1980s*

Constructionism differed from constructivism, as Papert (1980) posited an additional component to constructivism: students learned best when they were actively involved in *constructing* social objects. Using Taylor's (1980) tutor, tool, and tutee computer

analogy, Papert's constructionism advocates the tutee position. For example, the computer-as-tutee approach would involve students using Logo to teach the computer to draw a picture (Papert, 1980). Another technology example would be using another, slightly more advanced, microworld to teach Karel the Robot to perform various tasks.

### *Problem-Based Learning in the 1990s*

Although problem-based learning was developed in medical education in the 1950s, the methodology was not widely used in K–12 schools until the 1990s (Wilson, 1996). Problem-based learning involves students working on tasks and activities authentic to the environment in which those particular skills would be used. Students then learn by constructing their own knowledge from thinking critically and creatively to solve problems. This pedagogical practice caused a technological dilemma, in that desktop computers could not easily be transported around from place to place. Therefore, it is reasonable to claim that mobility became a desired attribute for technologies used with problem-based learning.

Learner-centered education, as the name implies, focuses on the role of the learner rather than the teacher; problem-based learning is a clear example of such a shift in the role of student and teacher. The teacher is the guide in the process, and no longer the main repository of knowledge (Hmelo-Silver, 2004). In the problem-based learning of the 1990s, students often worked in small groups of five or six to pool knowledge and resources to solve problems. This launched the start of the sociocultural revolution, focusing on learning in out-of-school contexts and the acquisition of knowledge through social interaction.

### *Socio-Constructivist Learning in the 1990s*

The next logical step in the learner-centered evolution was toward socio-constructivist learning. Social constructivists believe that social and individual processes are interdependent in the co-construction of knowledge (Sullivan-Palincsar, 1998; Vygotsky, 1978). The tenet of socio-constructivism is that intellectual advancement occurs through interactions with a group.

The sociocultural revolution was not limited to education specifically. SixDegrees.com was the first public social-networking site, launched in 1997 (Boyd & Ellison, 2007). This initial site developed into the plethora of social-networking sites available today, including Facebook, LinkedIn, and Twitter. Social networking sites provide "latent ties" (Haythornthwaite, 2005), which are those with established offline connections, and there is also the opportunity to meet online with people one may never meet face to face.

### *Learner-Centered Developments*

Thus far, a description has been given of the main learner-centered pedagogical developments from the 1970s to the end of the 1990s. There were other pedagogies/theories of learning during this time, such as discovery learning (Anthony, 1973), inquiry learning (Papert, 1980), and experiential learning (Kolb & Fry, 1975), which are similar to those described in this part. From studying the learning pedagogies and theories, it is clear that pedagogical practice since the 1970s has continually revised the model and theories behind learner-centered pedagogies. Table 1.1 provides a visual overview of this revision process.

**Table 1.1** Overview of the Revision Process in Learner-Centered Pedagogies/Theories

| Learner pedagogies/theories | Decade | Main tenets of the pedagogies/theories |
|---|---|---|
| Discovery learning | 1970s | Knowledge is discovered through active participation in the learning process |
| Constructivist learning | 1980s | Knowledge develops through interactions with the environment |
| Constructionist learning | 1980s | Knowledge is gained through actively creating social objects |
| Problem-based learning | 1990s | Knowledge is developed through working on tasks and skills authentic to the environment in which those particular skills would be used |
| Socio-constructivist learning | 1990s | Knowledge is co-constructed interdependently between the social and the individual |

The common evolving attributes listed in Table 1.1 are active involvement of the learner in the knowledge-making process and learner interaction with the environment and society. This is where we arrive at the learning pedagogies of the 2000s, with m-learning and context-aware ubiquitous learning providing new affordances for learners. M-learning and context-aware ubiquitous learning will be described later in this chapter.

Looking back at the brief descriptions of the technologies connected with the learning pedagogies of that time, it appears that technologies have had to play catch-up with pedagogical trends. However, there are those who believe that it is the technology leading pedagogical practice. Sharples (2005) proposed:

> Every era of technology has, to some extent, formed education in its own image. That is not to argue for the technological determinism of education, but rather that there is a mutually productive convergence between main technological influences on a culture and the contemporary educational theories and practices.
>
> (Sharples, 2005, p. 147)

The technological influence has also been reflected in educators and governments recently advocating for educational reforms to utilize technologies during teaching and learning (Common Core State Standards Initiative, 2010; Greenhow & Robelia, 2009; Jonassen, Howland, Marra, & Crismond, 2008). Teachers and students became increasingly aware of the potential to use the devices for differentiated, private, and self-directed learning.

Since the increase in technologies, such as the social adoption of the WWW and cell phones during the 1990s, the argument could be made that society's adoption and perceptions of technological hardware/software influence how people learn. The next section of this chapter gives a brief chronological overview of the technological underpinnings of technology relating to the development of m-learning.

## ADVANCES IN TECHNOLOGY

### *1970s*

The 1970s were a significant decade for the development of many hardware/software technologies such as the floppy disk, the microcomputer, the VHS videocassette recorder,

and the first mobile phone. This was also the decade in which Kay (1972) created the concept model of the Dynabook, the first handheld multimedia computer intended as a learning device. As Kay conceptualized the Dynabook, he described some of the attributes the revolutionary device would hold:

> Imagine having your own self-contained knowledge manipulator in a portable package the size and shape of an ordinary notebook. Suppose it had enough power to outrace your senses of sight and hearing, enough capacity to store for later retrieval thousands of page-equivalents of reference materials, poems, letters, recipes, records, drawings, animations, musical scores, waveforms, dynamic simulations, and anything else you would like to remember and change.
>
> (Kay & Goldberg, 1977/2001, p. 167)

The Dynabook was never actually created, but the ripples of his ideas have continued through to the 21st century. Sharples (2002) believed that the actual device did not move beyond the conceptual phase because technologies were not advancing fast enough. It also conflicted with the drive for the incorporation of desktop computers in classroom teaching.

Although the Dynabook never reached fruition, Kay and Goldberg's research developed prototype desktop computers, described as "interim Dynabooks" (Kay & Goldberg, 1977/2001, p. 168), and a programming language called SmallTalk. SmallTalk was an object-oriented software language, responsible for the later invention of the graphical user interface (GUI) for use on computers, portable media players, gaming devices, and handheld devices. This was a significant event in the history of electronic technologies, as the GUI allowed users to command the device through clicking on icons, rather than having to type in command strings, making it simpler for the novice user.

The first mobile phone, developed in 1973, was the Motorola DynaTAC 8000X, although it was not until 1983 that the commercial version was on sale. Nevertheless, the Dynabook and the first mobile phones paved the way for the m-learning devices that are readily available today.

### *1980s*

During this decade, numerous companies heralded the arrival of handheld computers. Table 1.2 lists a selection of the handheld devices introduced during the 1980s.

Other companies, such as Panasonic, Sharp, Texas Instruments, and Seiko Instruments, also produced commercial handheld computers in the 1980s. The handheld computers of the 1980s are still far from resembling the Dynabook conceived by Kay and Goldberg (1977/2001), and they were typically marketed and used in business settings, but a clear progression can be seen over the decade.

Technologies in general were becoming more personalized, moving from the shared desktop PCs to laptops and handheld personal computers, and from the fixed telephone to personal cell phones. During the early 1980s, the first commercial laptop computer was introduced to the market, and, during the late 1980s and 1990s, many schools and colleges began to allow students to bring laptops into schools and lecture halls. Cell phones also continued to evolve during the 1980s, becoming customizable, flexible, and increasingly smaller, with the phone connected to the person rather than the household (Goggin, 2006).

**Table 1.2** Selection of the Commercial Handheld Computers Introduced During the 1980s

| | | | |
|---|---|---|---|
| 1980 | TRS-80 Pocket Computer | Radio Shack | –24 × 1 text LCD display<br>–1.5K RAM |
| 1982 | PHC-8000 Handheld Computer | Sanyo | –One-line LCD display<br>–4kB RAM<br>–Allows connection to video monitor |
| 1982 | Pasopia Mini | Toshiba | –One-line LCD display<br>–4 kB RAM |
| 1984 | PB-700 Handheld Computer | Casio | –20 × 4 text LCD display<br>–4 kB RAM |
| 1989 | Portfolio, Portable Computer | Atari | –40 × 8 text LCD display<br>–128K, card slot for RAM<br>–MS-DOS-compatible computer<br>–Includes a speaker |
| 1989 | Poqet PC Computer | Poqet | –80 × 25 text LCD screen<br>–512 kB RAM<br>–Additional card slots for ROM or RAM |

From this trend toward personal technologies, educational establishments looked for a way to connect with the learner-centered approach evolving since the 1970s. Classroom response systems (CRSs) were just one such technology developed in the late 1980s as a way to reach the individual students in the classroom, although CRSs did not achieve widespread use until the late 1990s.

### *1990s*

During this decade, the learner-centered pedagogical movement was well established in schools, and, in a parallel development, technologies had become more advanced, personalized, and novice-friendly. Soloway, Guzdial, and Hay (1994) considered the direction in which technologies should progress, stating "Simply put, the HCI [Human Computer Interaction] community must make another transition: we must move from 'user-centered' design to 'learner-centered' design" (p. 38). Soloway et al. proposed that this should occur by considering three key questions: "Why support learners and learning? How might the interface support learners and learning? What are the issues involved in providing such support?" (p. 38). These were essential questions during a decade of significant technological adoption in the educational setting.

The 1990s were the decade in which the first digital camera, Web browser, and graphing calculator were developed. Many schools were using computer-assisted instruction programs on the multimedia computers. Palm Pilot personal digital assistants (PDAs) were the first multipurpose, handheld devices that could be utilized in the educational setting. The device ran basic programs, including calculator, tests, calendar, contacts, memos, photos, and notepad. In 1998, Sharples began an attempt at recreating the Dynabook. The Handheld Learning Resource (HandLeR) project studied the design of mobile devices in an attempt at creating an instrument to aid "lifelong learning," based on the tenets of experiential and collaborative learning (Sharples, Corlett, & Westmancott, 2001). Sharples et al. were also interested in learning contexts bridging formal and informal learning. The HandLeR was designed for use at any age or in any context, although the mentor character/animal interface was more suited for children than adults.

Portable digital devices have developed rapidly since they came onto the market in the latter part of the 1970s. From Kay's Dynabook concept in 1972, technologies have been progressing toward making Kay's dream a reality. Mobile devices have decreased in size and cost and increased in power, speed, memory, and functionality. The devices provide unique affordances for learner-centered pedagogies, which have further developed into the 21st century. The focus of this section has been directed toward the actual technological devices. One must also take into account the development of wireless wide-band technologies and application services—for example 3D phones, IEEE 802.11 (Wi-Fi) networks, 802.15.1 (Bluetooth) networks, active/passive radio-frequency identification, and global positioning system receivers—enabling an impressive system of networks for use on such devices (Caudill, 2007; Ding, 2010). Without such wireless technologies, m-learning would not exist. The next section of this chapter explicates the interconnectedness between the technologies and the learner pedagogies in the 21st century.

## CONNECTING THE TECHNOLOGIES AND THE LEARNING

In this chapter, m-learning has often been dichotomized into learning and technology so that we can better understand the changes that have taken place historically. However, the essence of m-learning is not in the learning or in the technology, but in the marriage between the two entities. This section explains the history of learning and technology as they became recognized as interconnected theories/pedagogies, namely e-learning and m-learning, while briefly touching upon a subdivision of m-learning titled context-aware ubiquitous learning (u-learning).

The first two sections of this chapter provide an overview of how e-learning and m-learning emerged from both a drive in educational philosophy and practice toward learner-centered pedagogies, and through technologies that put the learner at the heart of learning. These new methods of learning provide opportunities such as flexibility, accessibility, and convenience (Benedek, 2007), and a pedagogy that is personalized, learner centered, contextualized, and cooperative (Ding, 2010).

### *E-Learning*

Electronic learning appears to be the first recognized term to specifically connect learning with the technologies. Learning typically mimicked traditional teaching approaches, and early definitions describe e-learning as teaching and learning supported by electronic media and tools (e.g., Pinkwart, Hoppe, Milrad, & Perez, 2003). Researchers such as Keegan (2002) believed that e-learning was distance learning, which had been converted to e-learning through the use of technologies such as the WWW. As e-learning developed, along with the ever-expanding new technologies available for use, questions arose as to which electronic media and tools constituted e-learning; for example, did it matter if the learning took place through a networked technology, or was it simply learning with an electronic device? To clearly explain the technologies involved and to set e-learning apart from that of traditional learning, Tavangarian, Leypold, Nolting, and Voigt (2004) proposed that e-learning was:

> All forms of electronic supported learning and teaching, which are procedural in character and aim to effect the construction of knowledge with reference to individual

> experience, practice and knowledge of the learning. Information and communication systems, whether networked or not, serve as specific media to implement the learning process.
>
> (Tavangarian et al., 2004, p. 274)

The WWW made up a significant portion of the technological content fitting the category of e-learning during the 1990s and 2000s. Since society's adoption of the WWW in the 1990s, it has undergone radical changes. Websites have gone from static to dynamic and interactive, progressing from the read-only Web to the read–write Web (Richardson, 2005), which offered the users of the WWW more interaction and choice. In the past 20 years, a great many artifacts such as books, documents, and audiovisual materials were uploaded to the Web, as libraries and museums digitized collections, creating a bank of digital artifacts available to the public (Benedek, 2007). Those developing the content of the WWW utilized social theories of learning, offering social-networking sites and learning-management systems (LMSs) that were established and implemented in schools and universities across the Western world. LMS developed to mediate Web-based learning artifacts and communication between students and teachers.

Considering Table 1.1, which listed the tenets of learner-centered education, e-learning provides opportunities for learners to take advantage of many of the desirable learner-centered attributes. For example, students can be actively involved in the knowledge-making process, through writing a collaborative essay in Google Docs or defining a concept in a wiki. These examples show learners interacting with society, although the interactions are virtual, using computers based in a fixed location, with the learners shut off from the rest of the physical world. What e-learning lacked early on were physical interactions with the environment and society, without spatial and temporal limitations.

### *M-Learning*

During 2005, m-learning became a recognized term. M-learning as defined in this chapter encapsulates the attributes identified in Table 1.1 for learner-centered pedagogies. M-learning makes not only a step, but a leap further into the realm of learner-centered pedagogies. Still, this did not happen all at once.

Early on, m-learning was typically used to channel e-learning methods and techniques, quickly exposing the limitations of cell phones and PDAs compared with desktop computers at the time (Traxler, 2011). Early mobile technologies lacked functionality, screen size, processor speed, and battery life. Many of the unique opportunities offered by the mobile devices were not utilized.

As the 2000s progressed, the interest in PDAs decreased, as smartphones offered the same application and Web functionalities, but with the added mobile-phone capability. Cell phones, which were once a symbol of financial prowess, became a companion for the masses. Although technology was often an expensive option in higher education, colleges found that the number of students owning devices cut or abolished additional school cost entirely. Tablet computers continued the trend toward greater mobility. To put it simply, the first devices to be called tablets were laptops with a rotating screen and touch-screen capabilities, such as the Microsoft Tablet PC, commercially available in 2001. Ultra-mobile PCs, such as the Wibrain B1, were quietly introduced on the market in 2006 as smaller, more mobile versions of a laptop/tablet, but were quickly supplanted

by today's tablets, such as the iPad (2010) and Motorola Xoom (2011), which are overall more mobile that their initial counterparts.

As m-learning continued to develop, the multiple affordances the devices offered to further extend learner-centered pedagogies became evident. Traxler (2011) described five ways in which m-learning offers new learning opportunities: (1) contingent learning, allowing learners to respond and react to the environment and changing experiences; (2) situated learning, in which learning takes place in the surroundings applicable to the learning; (3) authentic learning, with the tasks directly related to the immediate learning goals; (4) context-aware learning, in which learning is informed by the history and the environment; and (5) personalized learning, customized for each unique learner in terms of abilities, interests, and preferences.

Technologies advanced to provide other forms of mobile technology, such as advanced tablets and laptops with many additional capabilities. Mobile phones now have the same capabilities as microcomputers, at a small fraction of the size. As m-learning is rapidly developing, offshoots of m-learning are being created. One such subdivision is context-aware ubiquitous learning, which describes learning that offers seamless services, adaptive services, and context-aware services (Yang, Zhang, & Chen, 2007), in which computing, communication, and sensor devices are integrated into the daily life of a learner. M-learning and the offshoot u-learning literally embody learner-centered education, in that learning will soon be omnipresent to the learner.

## CONCLUSION

Although m-learning is a relatively young field, this chapter provides a diachronic overview of how the philosophical, pedagogical, and conceptual underpinnings of a learner-centered approach, as well as the technological accomplishments, have engendered m-learning as the field it is today. Although this chapter has covered historical events leading to the field of m-learning, one must remember that this is merely one chapter of a book. For all the details that have been given, there are probably many that have been missed. A message to take away from this chapter is that, in 1972, Kay had a vision of a mobile-learning technology, compact enough to easily transport, with the capacity to store multiple reference materials and enough power to outrace the senses. Although the Dynabook was beyond the capabilities of technologies at that time, technologies have now caught up with this vision, and Kay's ideas for an alternative learning paradigm are appearing throughout the world.

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